

# Trait Fidelity in LLM-Based Multi-Agent Simulations: Model-Specific Behavioral Divergence in Commons Dilemma Scenarios

## Abstract

LLM agents are widely used in multi-agent social-dilemma simulations, but whether prompt-injected psychological traits produce behaviorally faithful outcomes is rarely tested with independent, code-computed metrics. We present AMADS (Academic Multi-Agent Decision Simulation), a LangGraph commons-dilemma framework where LLMs output structured extraction decisions and a deterministic, LLM-free referee computes all outcomes—eliminating circular LLM-as-judge evaluation. In `full_experiment_v1` (45 runs; 9 trait conditions  $\times$  5 replicates; Claude Haiku 4.5; Turkish prompts), `risk_tolerance_assigned` transfers reliably ( $r = +0.678$ ,  $p < 0.0001$ ), while `cooperation_assigned` exhibits inverse fidelity ( $r = +0.456$ ,  $p = 0.0016$ ): higher assigned cooperation predicts greater extraction. Trait-fidelity analysis, blind k-means clustering, and a formula-based control agent provide convergent validity. Eleven-language and eleven-trait pilot screens reveal no stable category rule for transfer; Sonnet 4.6 cross-model replication ( $n = 40$ ) reverses the cooperation pattern ( $r = -0.84$ ) while weakening risk ( $r = +0.15$ ). Trait fidelity is empirically idiosyncratic and model-specific—each trait-model-prompt combination requires validation before simulation use.

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## 1. Introduction

Commons dilemmas—in which self-interested extraction threatens a shared resource—have been studied for decades in behavioral and institutional economics (Walker, Gardner, & Ostrom, 1990). Recent work embeds LLM agents in similar settings to probe emergent cooperation, sustainability, and policy responses (Piatti et al., 2024). Parallel work at AAMAS 2025 explores trait-conditioned prompting for cooperative behavior in common-pool resource dilemmas (CFC-Prompt; Nguyen et al., 2025), treating natural-language trait descriptions as levers on agent decision-making. These studies assume that prompt-injected traits map onto measurable behavior, but validation is often indirect: outcomes are inferred from aggregate dialogue or LLM-as-judge scoring, raising circularity concerns.

The **trait injection problem** is central: when a researcher assigns `cooperation = 0.8` in a system prompt, does the agent extract less from a shared pool than when assigned `cooperation = 0.2`? Without independent, code-computed behavioral metrics, one cannot distinguish faithful trait transfer from concept misalignment, heuristic defaults, or model-specific reinterpretation of trait labels.

This paper makes three contributions:

1. **AMADS**, an open architecture for commons-dilemma multi-agent simulation with deterministic evaluation, parallel agent fan-out, and read-only agent views—so that every reported metric is computed without LLM involvement.
2. **Empirical trait-fidelity evidence** from a preregistered-style factorial experiment (`full_experiment_v1`,  $N = 45$ ) showing dissociated transfer for cooperation (inverse) versus risk tolerance (expected direction), validated by clustering and a formula-based control group.
3. **Generalization and cross-model replication** demonstrating that neither language nor trait category predicts fidelity, and that the same Turkish prompt yields opposite cooperation behavior on Claude Sonnet 4.6 versus Haiku 4.5—establishing trait fidelity as **model-specific**, not universal.

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## 2. AMADS Framework

### 2.1 Architecture

AMADS implements a commons-dilemma scenario as a LangGraph `StateGraph` with **parallel fan-out**: five agent nodes decide simultaneously each round without observing co-players’ current-round actions, followed by a single **Referee** node that collects decisions, updates pool dynamics, applies scheduled shocks, and computes metrics. Round progression is counter-driven (`round_number < max_rounds`); termination occurs on pool collapse or round completion.

```
Round Start      → Agent1 (isolated, parallel)
                  → Agent2 (isolated, parallel) → Referee (collect, compute, advance)
                  → Agent3...Agent5
```

Agents receive only an `AgentInputView`—their own frozen `TraitProfile`, the current `EnvironmentSnapshot`, and `round_number`—not the full `SimulationState`. This is enforced at the function-signature level, not merely in prompts.

### 2.2 Why an LLM-Free Referee

A recurring failure mode in LLM simulation research is **circular evaluation**: the same model (or another LLM) judges behavior it or a sibling model produced. AMADS separates decision generation from measurement entirely. The Referee performs deterministic math only—pool update, shock application, Gini computation, cooperation scoring, violation counting—and **never calls an LLM**. LLMs produce structured `AgentDecision` records; all inferential statistics derive from numeric fields computed in code.

## 2.3 State Structure and AgentDecision Schema

State updates flow through Pydantic v2 models in `core/state.py`:

| Layer         | Key types                                                       | Write policy               |
|---------------|-----------------------------------------------------------------|----------------------------|
| Frozen inputs | <code>TraitProfile</code> ,<br><code>ShockEvent</code> schedule | Set at init; never mutated |
| Agent output  | <code>AgentDecision</code><br>(appended per round)              | Agent nodes append only    |
| Environment   | <code>EnvironmentSnapshot</code>                                | Referee only               |
| Metrics       | <code>MetricsSnapshot</code><br>(history list)                  | Referee only               |

Each round, an agent outputs:

```
class AgentDecision(BaseModel):
    agent_id: str
    round_number: int
    extraction_amount: float = Field(ge=0.0)
    justification: str = Field(max_length=500) # logging only
    declared_max: float = Field(ge=0.0)
```

`justification` is excluded from all metric formulas. Observed cooperation is computed as  $1 - (\text{extraction} / \text{max\_extractable})$ ; observed risk-related behavior uses extraction variance in post-shock rounds. **Trait-behavior fidelity** is Pearson  $r$  between assigned trait values and these code-derived quantities.

Pool dynamics follow a fixed recurrence:

```
pool[t+1] = clamp(pool[t] - Σ extraction_i[t], 0, capacity) ×
regen_rate.
```

## 3. Experimental Design

### 3.1 Primary Experiment: `full_experiment_v1`

| Parameter                | Value                                                                       |
|--------------------------|-----------------------------------------------------------------------------|
| Traits                   | <code>cooperation_assigned</code> ,<br><code>risk_tolerance_assigned</code> |
| Levels per trait         | Low = 0.2, Medium = 0.5, High = 0.8                                         |
| Conditions               | $3 \times 3 = 9$ (full factorial)                                           |
| Replicates per condition | <b>5</b> → <b>45 runs</b> total                                             |
| Agents per run           | 5 (homogeneous: identical trait pair for all agents in a condition)         |

| Parameter       | Value                                                                    |
|-----------------|--------------------------------------------------------------------------|
| Rounds          | 15; shock at round 7 (CAPACITY_DROP, −20%)                               |
| Model           | claude-haiku-4-5-20251001                                                |
| Prompt language | Turkish (trait field names and structured output keys in English)        |
| Outcome         | 45/45 runs finished; all terminated via resource collapse (rounds 10–15) |

Trait values were injected via a locked symmetric system prompt (`agents/decision_agent.py`):

*“Sen bir ortak kaynak havuzundan çekim yapan bir agentsin. cooperation\_assigned değerin {cooperation\_assigned:.2f} (0=tamamen bencil, 1=tamamen işbirlikçi), risk\_tolerance\_assigned değerin {risk\_tolerance\_assigned:.2f} (0=çok temkinli, 1=çok risk alan). Kararını bu eğilimlere uygun ver, ama bu sayıları çıktıda tekrar etme veya açıklama.”*

Primary fidelity analyses use **round 0** extraction fractions to avoid confounding with collapse timing (round-averaged cooperation scores correlate  $r = 0.95$  with `collapse_round`).

### 3.2 Calibration Decisions

| Parameter              | Locked value | Rationale (one sentence)                                                                                                                                  |
|------------------------|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| EXTRACTION_LIMIT_RATIO | <b>0.12</b>  | Real-LLM calibration showed that 0.12 guarantees shock exposure at round 7 while preventing spurious 15-round completion under aggressive extraction.     |
| COLLAPSE_EPSILON_RATIO | <b>0.01</b>  | Float pool values asymptotically approach but never reach exact zero; treating pool 1% of capacity as collapse fixes misclassified “completed” dead runs. |

| Parameter                | Locked value | Rationale (one sentence)                                                                                                                                                                               |
|--------------------------|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>temperature</code> | <b>0.2</b>   | Fixed across all runs to reduce stochastic variance without claiming full reproducibility; observed Gini variance ( $\sigma^2 = 4.78 \times 10^{-4}$ ) already provided sufficient between-run signal. |

Mock-agent calibration at ratio 0.30 appeared viable but failed on real LLMs (0% round-7 shock exposure), underscoring that parameter tuning must be confirmed with actual model calls.

### 3.3 Power Analysis

Using  $\alpha = 0.05$  as a uniform minimum detectable difference:

- **Gini coefficient** ( $\sigma^2 = 0.0197$ ): corresponds to a large standardized effect ( $d = 2.5$ );  $N = 45$  is **more than sufficient**.
- **Cooperation score average** ( $\sigma^2 = 0.108$ ): corresponds to a small effect ( $d = 0.46$ );  $N = 45$  is **insufficient** ( $\sim 75$  replicates per condition would have been needed for conventional power).

We therefore report cooperation-related inferential results with effect sizes and  $p$ -values while noting limited power for condition-level mean comparisons on cooperation score averages.

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## 4. Primary Findings — Haiku

### 4.1 Cooperation: Inverse Fidelity

On round-0 extraction fractions ( $n = 45$ ):

- **Cooperation  $\rightarrow$  extraction\_fraction**:  $r = +0.456$ ,  $p = 0.0016$  (positive correlation = **inverse fidelity** relative to the experimental operationalization, where higher cooperation should predict *lower* extraction).

Marginal pattern (micro-A/B, risk fixed at 0.2): **cooperation** = 0.2  $\rightarrow$  mean extraction = 4.56; **cooperation** = 0.8  $\rightarrow$  mean extraction = 8.40. The effect persists when risk is held constant, ruling out a risk $\times$ cooperation interaction artifact.

An initial round-averaged cooperation score showed  $r = -0.345$  ( $p = 0.02$ ) but was confounded with collapse timing; round-0 unconfounded metrics supersede

it.

## 4.2 Risk Tolerance: Expected Direction

- **Risk**  $\rightarrow$  **extraction\_fraction**:  $r = +0.678$ ,  $p < 0.0001$  ( $n = 45$ ).

High assigned risk reliably predicts higher extraction fractions; no concept inversion was observed for this trait.

## 4.3 Convergent Validity (Three Independent Methods)

| Method                                                            | Risk                                                  | Cooperation                                               |
|-------------------------------------------------------------------|-------------------------------------------------------|-----------------------------------------------------------|
| <b>Trait fidelity</b> (Pearson $r$ , round 0)                     | $r = 0.678$ , $p < 0.0001$                            | $r = +0.456$ , $p = 0.0016$ (inverse)                     |
| <b>K-means clustering</b> ( $k = 2$ , $n = 45$ , blind to traits) | Partial overlap (ARI vs. risk level = <b>0.2862</b> ) | Does not separate (ARI vs. coop level = <b>0.1235</b> )   |
| <b>Control group</b> (deterministic formula agent, $n = 27$ )     | N/A (risk not implemented in control)                 | $r = -1.000$ (expected); LLM $r = +0.456$ (opposite sign) |

*Clustering run (analysis/clustering.py, full\_experiment\_v1): silhouette-selected  $k^* = 2$ ; combined 9-cell ARI = 0.1469 (low overall overlap).\**

The deterministic control agent implements the design-intended mapping; mock agents in code likewise yield cooperation  $r = +1.0$ . Only real LLM agents invert cooperation—confirming the finding is behavioral, not implementation error. Prompt symmetry between cooperation and risk sentences was verified character-by-character.

## 4.4 Concept Misalignment Mechanism

Qualitative review of **justification** texts (logging-only, not used in metrics) indicates the model interprets “cooperation” not as “extract less to preserve the commons” but as “**use one’s own share responsibly and sustainably.**” Representative paraphrases from agent outputs include:

- Framing high extraction ( $\sim 70\%$  of **declared\_max**) as “*ensuring other agents also benefit.*”
- Describing aggressive withdrawal as “*supporting sustainability*” of the shared pool.

Under this frame, a “cooperative” agent fulfills duty by extracting decisively yet “fairly,” which aligns with **higher** numeric extraction—not lower—relative to our operational definition ( $1 - \text{extraction}/\text{max\_extractable}$ ).

## 5. Generalization

### 5.1 Eleven-Language Pilot ( $n = 5$ per cell per language; 110 calls total)

To test whether inverse cooperation fidelity is a Turkish translation artifact, we ran identical micro-A/B screens (risk = 0.2 fixed; cooperation = {0.2, 0.8}) in 11 languages with Haiku 4.5. Classification used  $|\text{difference}| \geq 0.30$  between high- and low-cooperation mean extraction.

| Language | coop=0.2<br>avg.              | coop=0.8<br>avg.         | Difference                    | Class                    |
|----------|-------------------------------|--------------------------|-------------------------------|--------------------------|
| tr       | 4.80                          | 8.40                     | +3.60                         | inverse<br>fidelity      |
| zh       | 8.40                          | 9.60                     | +1.20                         | inverse<br>fidelity      |
| es       | 8.94                          | 9.60                     | +0.66                         | inverse<br>fidelity      |
| en       | 9.12                          | 9.60                     | +0.48                         | inverse<br>fidelity      |
| ja       | 9.12                          | 9.60                     | +0.48                         | inverse<br>fidelity      |
| hi       | 9.16                          | 9.60                     | +0.44                         | inverse<br>fidelity      |
| fr       | 9.76                          | 9.60                     | -0.16                         | trait-blind              |
| ru       | 9.60                          | 9.60                     | 0.00                          | trait-blind              |
| de       | 8.16 $\rightarrow$ 8.88*      | 7.68 $\rightarrow$ 9.12* | -0.48 $\rightarrow$<br>-0.24* | trait-blind at<br>$n=10$ |
| pt       | 9.60 $\rightarrow$ 9.36*      | 9.12 $\rightarrow$ 9.60* | -0.48 $\rightarrow$<br>-0.24* | trait-blind at<br>$n=10$ |
| ar       | 10.30 $\rightarrow$<br>10.45* | 9.60 $\rightarrow$ 9.85* | -0.70 $\rightarrow$<br>-0.60* | borderline<br>expected   |

\*Extended validation ( $n = 10/\text{cell}$ ) for de, pt, ar.

**Finding:** Turkish shows the **strongest and most consistent** inverse fidelity (+3.60)—not a translation bug but the clearest signal. Six additional languages show weak inverse direction; two are trait-blind. Initial “expected direction” labels for German and Portuguese collapse to trait-blind at  $n = 10$ . **No language yields reliable expected-direction cooperation fidelity** at pilot scale.

### 5.2 Eleven-Trait Pilot (~100 calls; English prompts)

Nine candidate traits beyond cooperation and risk were screened (all others fixed at 0.5; target trait = {0.2, 0.8};  $n = 5/\text{cell}$ ).

| Class                                   | Traits                                                                                                                                            | Evidence                                    |
|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|
| <b>Working</b> (4)                      | <b>risk_tolerance</b> ,<br><b>aggression</b> , <b>fairness</b> ,<br><b>greed</b>                                                                  | Numeric separation in<br>expected direction |
| <b>Unreliable /<br/>trait-blind</b> (7) | <b>cooperation</b> (inverse),<br><b>impatience</b> , <b>tolerance</b> ,<br><b>creativity</b> , <b>hoarding</b> ,<br><b>trust</b> , <b>caution</b> | No separation or wrong<br>direction         |

**No category rule holds:** early hypothesis that action-oriented traits transfer reliably was **refuted** by **caution\_assigned**—defined as the inverse frame of risk (0 = not cautious, 1 = very cautious)—which produced **trait-blind** behavior (fixed extraction 6.0 at both levels) while **risk\_tolerance** worked. Word choice and model heuristics dominate over abstract/concrete or action/value taxonomy.

*Note:* **hoarding** failure may reflect single-round design (stockpiling requires multi-round accumulation) rather than pure trait blindness.

## 6. Cross-Model Replication

### 6.1 Sonnet 4.6 Pilot (**sonnet\_crossmodel\_v1**, $n = 40$ )

| Parameter | Value                                                                  |
|-----------|------------------------------------------------------------------------|
| Model     | <b>claude-sonnet-4-6</b>                                               |
| Prompt    | <b>Identical Turkish text</b> to Haiku experiments                     |
| Grid      | cooperation $\times$ risk {0.2, 0.8} <sup>2</sup> , 10 replicates/cell |
| Round     | 0 only (single-round, no collapse confound)                            |

| Trait                                   | Sonnet 4.6 ( $n = 40$ )                  | Haiku 4.5 (round 0, $n = 45$ )         |
|-----------------------------------------|------------------------------------------|----------------------------------------|
| Cooperation $\rightarrow$<br>extraction | $r$ <b>−0.84</b> (expected<br>direction) | $r$ <b>+0.46</b> (inverse<br>fidelity) |
| Risk $\rightarrow$ extraction           | $r$ <b>+0.15</b> (weak)                  | $r$ <b>+0.68</b> (strong)              |

Sonnet marginal means (cooperation): low (0.2) 10.0 extraction vs. high (0.8) 5.35—high cooperation predicts **less** extraction, opposite to Haiku.



## 6.2 Haiku vs. Sonnet Comparison

| Dimension            | Haiku 4.5                               | Sonnet 4.6                                        |
|----------------------|-----------------------------------------|---------------------------------------------------|
| Cooperation fidelity | Inverse ( $r = +0.46$ )                 | Expected ( $r = -0.84$ )                          |
| Risk fidelity        | Strong ( $r = +0.68$ )                  | Weak ( $r = +0.15$ )                              |
| Same prompt          | Turkish, locked                         | Turkish, identical                                |
| Interpretation       | Concept misalignment<br>for cooperation | Cooperation aligns with<br>operational definition |

**Primary cross-model contribution:** Trait fidelity is **model-specific**. The same trait labels, prompt template, and scenario can produce opposite behavioral signatures across models—a pattern invisible when studies report results from a single model family.

Whether the apparent cooperation–risk “trade-off” (strong risk / weak-inverse cooperation on Haiku vs. the reverse on Sonnet) is systematic or coincidental **cannot be determined with  $N = 2$  models**.

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## 7. Discussion

### 7.1 Practical Recommendation

Researchers should treat every candidate trait—and every model–trait–prompt combination—as **hypothesis requiring empirical micro-validation** (single-round A/B extraction screens) before committing to full factorial experiments. Category intuition (abstract vs. concrete, action vs. value, synonym pairs like risk/caution) does not predict transfer success. AMADS follows the human CPR literature in treating numeric extraction from a shared pool as the primary behavioral endpoint (Nockur et al., 2023), but our results show that assigning a cooperation label in prompt does not guarantee alignment with that operationalization in LLM agents.

### 7.2 Limitations

- **Models:** Primary data from one model (Haiku 4.5); cross-model evidence from one additional model (Sonnet 4.6) only.
- **Scenario:** Single commons-dilemma formulation; generalization to bargaining or crisis scenarios is untested.
- **Agent design:** Homogeneous trait assignment within each run (all five agents share the same profile); heterogeneous populations may differ.
- **Language:** Main experiment used Turkish prompts; multilingual pilots were short-screen pilots, not full 45-run replications.
- **Power:** Cooperation score averages underpowered at  $N = 5$ /condition; round-0 fidelity metrics are primary inferential evidence.

- **Shock analysis:** Post-shock risk operationalization deferred; round-0 findings are the locked primary result.

### 7.3 Transparency: `experiment_conditions` Migration

Run-to-trait mappings for `full_experiment_v1` and `control_group_v1` are stored in an `experiment_conditions` database table that underwent a post-hoc schema migration (`core/database.py`) to normalize condition identifiers; all analyses join through this table and remain consistent with locked findings.

### 7.4 Future Work

- **Cross-model panel:** Replicate the full  $9 \times 5$  factorial on 3 models (e.g., Opus, GPT-4 class) under preregistered analysis plans to test whether cooperation–risk trade-offs are systematic.
- **Heterogeneous agent design:** Assign mixed trait profiles within runs to test whether homogeneous conditioning amplifies misalignment.
- **Prompt revision experiments:** Behavioral rewording of cooperation (e.g., “0 = extract maximally, 1 = extract minimally”) as a **separate** `experiment_id`—not retroactive relabeling of `full_experiment_v1`.

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## 8. Conclusion

We built AMADS to measure LLM agent behavior in commons dilemmas through deterministic, non-circular evaluation, and ran a 45-run factorial experiment plus convergent validation, multilingual/trait pilots, and cross-model replication. Haiku 4.5 shows reliable risk transfer but inverse cooperation fidelity driven by concept misalignment; neither language nor trait category predicts success, and Sonnet 4.6 reverses the cooperation pattern under the same prompt. We recommend empirical micro-validation of every trait–model pair before using LLM agents as stand-ins for psychologically profiled populations in simulation research.

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## References

(DOIs provided where available; *arXiv preprints noted*.)

- Piatti, G., Rane, A., Shi, W., Hofstätter, F., Li, B., Gao, Y., Bernstein, A., & Sumita, E. (2024). Cooperate or Collapse: Emergence of Sustainability Behaviors in a Society of LLM Agents. *arXiv:2404.16698*.
- Nguyen, D., Le, H., Do, K., Gupta, S., Venkatesh, S., & Tran, T. (2025). Navigating Social Dilemmas with LLM-based Agents via Consideration of Future Consequences: Extended Abstract. In *Proceedings of the 24th*

*International Conference on Autonomous Agents and Multiagent Systems (AAMAS 2025)* (pp. 2693–2695). IFAAMAS.

- Hardin, G. (1968). The tragedy of the commons. *Science*, 162(3859), 1243–1248.
- Nockur, L., Pfattheicher, S., & Keller, J. (2023). From asymmetric to symmetric consumption opportunities: Extractions from common resources by privileged and underprivileged group members. *Group Processes & Intergroup Relations*, 26(8), 1819–1840. <https://doi.org/10.1177/13684302221132722>
- Ostrom, E. (1990). *Governing the Commons: The Evolution of Institutions for Collective Action*. Cambridge University Press.
- Walker, J. M., Gardner, R., & Ostrom, E. (1990). Rent dissipation in a limited-access common-pool resource: Experimental evidence. *Journal of Environmental Economics and Management*, 19(3), 203–211. [https://doi.org/10.1016/0095-0696\(90\)90008-2](https://doi.org/10.1016/0095-0696(90)90008-2)